

How Much Do You Know About the Environmental Impact of AI-Generated Images?

Posted by u/cisfer • 0 points • 26 comments

Hey everyone!

I'm conducting a research project on the **environmental impact of AI-generated images**—specifically in the context of digital design—and I'd love to hear from you! The goal is to understand how designers and creatives use these tools and how aware we are of their hidden environmental costs.

If you're a **web designer, digital artist, or creative professional**, I'd greatly appreciate your input. The survey is short and available in **English and Portuguese**. Your responses will help shed light on an often-overlooked topic.

[Take the survey here](<https://form.jotform.com/250673632253051>)

Thanks for your time, and feel free to share with others who might be interested!

Comments

u/Back_Up_For_Block • 1 points • 2025-03-27 01:06 UTC

u/bot-sleuth-bot

u/bot-sleuth-bot • 1 points • 2025-03-27 01:07 UTC

Analyzing user profile...

Account has used the same title for multiple posts on multiple subreddits.

One or more of the hidden checks performed tested positive.

Suspicion Quotient: 0.47

This account exhibits a few minor traits commonly found in karma farming bots. It is possible that u/cisfer is a bot, but it's more likely they are just a human who suffers from severe NPC syndrome.

^(I am a bot. This action was performed automatically. Check my profile for more information.)

u/diogodiogod • 3 points • 2025-03-26 21:45 UTC

"hidden environmental costs" is a catchphrase seen very badly by most people on the open community (IMO). Base models sure use a lot of power to be created, but after that, it is done...

the power to generate images locally is insignificant... barely none compared to playing a game for hours and hours.

Are you also accounting the "hidden environmental costs" of using photoshop? Developing a game asset? Playing games? Rendering an animation for a film?...

IDK. I hate this "hidden environmental costs" argument. Everything have a "hidden environmental costs" and this is no exception. A lot of other yechinology and "art" fields have way more "hidden environmental costs" that no one is talking about. But they need a new argument against this new tech, I guess...

u/beti88 • 5 points • 2025-03-26 21:00 UTC

The same energy cost as playing games for the same amount of time. Cuz I use my own computer. I think 90% of people here also do the very same

u/NetworkSpecial3268 • -7 points • 2025-03-26 20:48 UTC

Waste of time... Never gonna get a thoughtful answer here.

u/NetworkSpecial3268 • -6 points • 2025-03-26 20:48 UTC

Waste of time... Never gonna get a thoughtful answer here.

u/Lucaspittol • 3 points • 2025-03-26 20:26 UTC

China is opening a new coal power plant pretty much every week. Even if everyone here generates AI images locally, even considering model training, it is a drop in the bucket.

Note: USP is the university making the survey, they are notoriously left-wing in Brazil.

u/cisfer • 1 points • 2025-03-26 20:41 UTC

Thank you for the observation. I am a student at USP, but the research is mine as part of my MBA studies at USP.

u/pls_pm_me_your_tits8 • 12 points • 2025-03-26 20:19 UTC

This is a bit of a weird post. As far as I know most people in this sub generate images locally so taking that into consideration the question of the thread would revolve around how clean is the electricity generated in the user country of residence

u/Lucaspittol • 4 points • 2025-03-26 20:30 UTC

China and India are burning coal like no tomorrow and people in the West are "concerned" about the use of electricity to run a GPU locally, which is basically nothing. OP lives in Brazill, which already generates like 80% or more energy from clean sources (mostly hydro, wind, solar and a small amount from nuclear).

u/Vaughn • 8 points • 2025-03-26 20:19 UTC

Judging from the contents of the survey, it seems you'd benefit from a follow-up survey to inform you about how much energy image generation uses. I've rarely seen a more biased set of options.

u/cisfer • 1 points • 2025-03-26 20:46 UTC

Thank you for the observations. I would be really interested in understanding which points you believe are biased so I can attempt a correction?

u/mcmonkey4eva • 1 points • 2025-03-28 23:53 UTC

Local AI image generation uses a fraction of 1 watt-hour of energy per image, a smartphone has 10+ watt-hours of battery, so your survey's scale of "how much do you think it uses" that starts at "a quarter of a smartphone charge" and goes up to two charges, is an invalid question leading towards extremely wrong answers. You seem to have been misled by the "make up a terrifyingly massive number" crowd that also reports every ChatGPT response uses 600 gallons of water or whatever nonsense like that. It just isn't true at all.

u/KoenBril • 3 points • 2025-03-26 22:11 UTC

The fact that you don't know how AI image generation works, but seem to have concluded that there are "hidden costs" involved in it is a bias in itself.

Start by educating yourself in the technology, before you suggest that the people using it don't know the costs involved.

u/PVPicker • 8 points • 2025-03-26 20:17 UTC

I run it locally on repurposed/refurbished hardware that's running off of solar power that otherwise would go to waste. Zero environmental impact.

Also, most of these studies are flawed because they're ignoring the environmental impact of a human using a computer for 20+ hours ($100W \times 20 = 200$ watt hours) to accomplish the same thing as a 3090 using 400W can in 10 seconds while pulling 400W (1 watt hour).

u/cisfer • 1 points • 2025-03-26 20:40 UTC

Thank you for the observations. My plan it to also include the counter narrative to AI costs through initiatives and approaches such as yours, at the next phase of the research when I am writing the thesis.

u/PVPicker • 3 points • 2025-03-26 22:31 UTC

Your thesis is seemingly flawed as you're attempting to present a view instead of the truth. Power hungry applications such as AI, crypto, etc often "follow the energy" and go wherever it's cheapest. Solar and renewables are incredibly cheap in terms of cost per kwh, and are being built up to meet this demand. Coal/other sources are bad for AI as they cost significantly more.

It seems like you found some studies about energy usage of AI, and wanted to see if people are aware of it.. It feels like you you came here to almost 'preach' and see how much we knew about energy usage of AI image generation. Meanwhile you've gotten 'owned' in the comments. You're going to need to do a 180 on all this as based on your questioning I don't think your thesis is defensible. AI has high energy demands, but it's still far far far less than the energy demands of a human + computer doing it on it's own. That's why AI is so popular.

u/size12shoebacca • 10 points • 2025-03-26 20:14 UTC

None of your options reflect people running models locally, which is what most of this community is. Just figured I'd explain why your post is met with downvotes here.

u/cisfer • 2 points • 2025-03-26 20:36 UTC

Thank you for the insight. I am really trying to understand more about the topic, hence the research.

u/size12shoebacca • 1 points • 2025-03-26 21:12 UTC

A lot of us run models locally that are watered down versions of the big models, that are still fantastically powerful but run on consumer gaming hardware.